PROJECT CCHAIN | SURVEILLANCE MODEL RESULTS

Executive Summary & Empirical Performance Report

Target City: Cagayan de Oro (80 Barangays, 2003-2022 Data)

1. Executive Project Summary

* **Dataset Source:** Official Project CCHAIN Kaggle Dataset (thinkdatasci/project-cchain)
* **Consortium Origin:** Thinking Machines, EpiMetrics, Manila Observatory, PACSII (Wellcome Trust / Lacuna Fund)
* **Target Region:** City of Cagayan de Oro (PH104305000) across all 80 Barangays
* **Temporal Scope:** 20 Years of Data (January 2003 - December 2022) at Monthly Grain
* **Target Disease:** Dengue Fever (ICD-10 Code: A90-A91)
* **Final Data Scale:** 18,960 monthly barangay rows x 37 aligned feature columns (9.18 MB CSV)

2. Machine Learning Model Architecture & Prediction Task

The model (Random Forest Classifier with 150 decision trees) analyzes weather conditions from 1, 2, and 3 months ago combined with urban building density and population numbers to predict whether a barangay will experience a Dengue Outbreak in the upcoming month.

- Outbreak Definition: Monthly cases >= 75th historical percentile of that specific barangay.
- Train Set: 2003-2018 (15,120 historical rows used to learn patterns).
- Test Set: 2019-2022 (3,840 unseen future rows used for evaluation).

3. Numerical Model Performance Results

Evaluation Metric	Result Score	Epidemiological Interpretation
ROC-AUC Score	0.903 (90.3%)	Outstanding ability to rank high-risk outbreak periods over normal background levels.
Overall Accuracy	87.0%	3,340 out of 3,840 monthly barangay test evaluations predicted correctly.
Outbreak Recall	72.0%	Successfully catches 72% of all actual outbreaks 30 to 60 days before hospital spikes.

4. Top Predictive Climate-Health Indicators (Feature Importances)

Indicator Feature	Importance	Real-World Biological Rationale
pop_density_mean	35.05%	High human host concentration enables rapid vector-to-host virus transmission.
google_bldgs_pct_built_up_area	14.92%	Impervious urban surfaces create artificial container breeding sites for mosquitoes.
heat_index_mean_lag_3m	9.39%	3-month heat stress accelerates virus incubation (EIP) inside mosquitoes.
heat_index_mean_lag_2m	6.93%	2-month ambient heat elevation speeds up mosquito larval hatching.
tave_mean_lag_1m	5.72%	1-month mean temperature triggers increased mosquito biting frequency.

5. Prescriptive Action Plan for Local Health Offices (LGUs)

Outbreak Risk Level	Probability (P)	Automated Prescriptive Response
Level 1: Normal	P < 0.30	Standard community sanitation & routine vector index monitoring.
Level 2: Alert	0.30 <= P < 0.65	Pre-emptive larviciding in high-density barangays; dispatch BHW workers.
Level 3: Outbreak Warning	P >= 0.65	Immediate targeted spatial fogging, reserve hospital triage beds, deliver IV fluids.